

Project 08 — How Coupling Strength Changes Network Dynamics

Time to finish: 1 day (about 4-6 hours, working alone or in a pair).

What you will do

You are given the activity of 5 coupled nodes at several coupling strengths w . Your job is to measure how strongly each node's activity fluctuates (its standard deviation) and see how that changes with w .

Background

In a linear network, the activity amplitude depends on the coupling strength w . As w gets close to a critical value, the amplitude grows sharply — a sign of an approaching instability (bifurcation).

Data

In `shared/data/`:

- `fit_timeseries.csv` — activity of nodes $x_0..x_4$ for several w values (subthreshold).
- `holdout_timeseries.csv` — same, at near-threshold w (for a bonus check).

Steps

1. Load and look. Open the starter notebook. Load `fit_timeseries.csv`. Print the unique w values and shape.
2. Plot one time series. Pick one w and plot $x_0(t)$, $x_1(t)$, $x_2(t)$, $x_3(t)$, $x_4(t)$ on one figure.
3. Amplitude vs w . For each w , compute the standard deviation of each node's activity. Take the mean across nodes. Plot amplitude vs w .
4. Check the holdout. Repeat step 3 for the holdout data. Does the amplitude grow more sharply as w gets larger in magnitude?

What to hand in

- Your filled notebook.
- The time-series plot from Step 2.
- The amplitude-vs- w plot for fit data.
- The amplitude-vs- w plot including holdout data.

- A short paragraph (3-4 sentences) on how the network responds to stronger coupling.

If you finish early: for one w , plot the correlation matrix between the 5 nodes. Which nodes are most correlated?